

FREDERICK JEFFCOCK

Internship - March 2027 - Aug 2027

SOFTWARE ENGINEERING STUDENT

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TECHNICAL SKILLS

- **Languages:** C • C++ • Python
- **Tools:** Git • GitHub • GDB

PROJECTS

Raytracer [Uni] [C++]

April 2026

- Modular raytracing engine (3D rendering) based on an extensible architecture.
- Implementation of intersection calculations, lighting, shadows and reflections
- Plugin system via dynamic library loading (.so)
- Implemented multi-threading optimization that slashed rendering times by 50% to 90% depending on the size of the image.

Cartridge [C]

April 2026

- Low level development of a set of games that runs on Game Boy hardware.
- Implementation of an engine that interfaces with the hardware.

CHIP8 Interpreter [C]

June 2026

- Implemented a CHIP-8 virtual machine, reproducing its instruction set and execution model.
- Implemented the fetch/decode/execute cycle, virtual registers and memory.

RISC-V32i Emulator (WIP) [C]

August - 2026

- Implementing the RISC-V RV32I unprivileged instruction set.
- CPU fetch/decode cycle, virtual registers and memory.

PROFESSIONAL EXPERIENCE

Full-stack Intern, Ir2 Cabinet

Oct 2025 - Dec 2025

- **Web Scraper development | Python** - scraper targeting job listing aggregates like LinkedIn/ FranceTravail through HTML scraping and API calls.
- Dashboard Interface - displays the results of the scraper in readable manner.
- SQLite database - to store the results and for easy sharing.

Full-stack Intern, Kilanne Investments

June 2024 - Aug 2024

- Maintained old systems.
- Web Scraper | **Rust** - Targeting market data.

EDUCATION

EPITECH - Software Engineering

Project-based engineering program

University Montpellier - Economics

Economics undergrad

Sep 2024 - June 2029

Sep 2021 - Sep 2023